1. **Simplify** the following boolean expression: $(A + B)(C+B)(C+A)$

2. **Find** the complement of the following expression:
   $(x + y' + z')(x' + y)(x' + z')$

3. **Draw** the following functions using NOR gates only:
   $F(A,B,C,D) = (A'BCD' + A'C' + A(B'+D'))$
   NB: You can't simplify the above function. You have to draw based on the function given in question.

4. **Draw** the following functions using NAND gates only:
   $F(A,B,C,D) = (A'B'+C'D + CD + (B'C+D'))$
   NB: You can't simplify the above function. You have to draw based on the function given in question.

Find out SOP and POS for the following: (Do not use truth table)

5. $F(A,B,C) = A'C+BC'$
6. $F(A,B,C,D) = A+ B'C'D$
7. $F(A,B,C,D,E) = ACE+CDE$